

Technology has become integrated into our daily lives. We use various electronic devices to perform many tasks. We use them to communicate with people, listen to music, browse the internet, complete our assignments, respond to emails, and many other tasks. Most people own more than one device and with these devices comes the issue of having to replace the device once it becomes slower, decreases in battery life, runs out of storage, stops getting software updates, gets damaged, or is no longer functional. But what do most people do with the old device once they upgrade to the new one? Some people sell their device or recycle their device. Those are great solutions since the device could be sold to get some extra cash and it reduces waste. The device could be recycled to conserve valuable resources that are used to create circuit board parts, connectors, and wiring. There are a lot of people who end up throwing away their old electronic devices. This solution may seem harmless, but it creates some big issues. Developed countries like the US offload tons of electronic waste into developing countries causing serious health and environmental issues.

If one person tosses their electronic device then it is no big deal since it is just one device, but it starts to become an issue when millions of devices are thrown away each year. For example, “the US generates approximately 6.9 million tons of e waste each year and only 17.4% of this waste is being recycled properly” (Smith, 2023). Globally the number is much bigger, “The Global E-waste Monitor estimated that 53.6 million metric tons (Mt) of e-waste were produced globally in 2019. This figure is projected to grow to 74.7 Mt by 2030” (Parvez, 2021).

With so much e waste many developed countries have no choice but to export this waste to other countries. Many developed countries tend to export e waste to developing countries like China, India, and African countries. They send the e waste to these countries because labor costs tend to be cheaper and there are less environmental regulations in place. The stakeholders that we will focus on for this case study are Electronics Manufacturers, the federal Government, and consumers.

The stakeholders all have a crucial role in influencing this problem. Electronics Manufacturers are responsible for designing and producing consumer electronics. They can directly influence how long a product can last and how a product can be repaired. Their motivation is to increase profit often at the expense of environmental sustainability. The federal government is responsible for creating and enforcing laws about e waste exports. For example, the US has not signed the Basel convention that focuses on limiting the amount of e waste that is exported. Their motivation is political pressure and trade relationships. Consumers are not directly responsible for e waste exports, but they indirectly contribute to this problem because of their limited knowledge of where their discarded electronics end up. Their motivation is cost, convenience, and desire to upgrade to newer technologies.

To analyze the ethical dimensions of exporting e-waste from developed countries like the US to developing countries, I will apply two ethical frameworks: Deontological ethics and Virtue Ethics. I will also reference professional engineering codes of ethics IEEE and NSPE to evaluate the responsibilities of engineers. Deontological Ethics evaluates making decisions based on moral rules or duties. This framework will assess whether it is ethically acceptable to send tons of hazardous e waste to countries that are not able to manage it safely. It emphasizes the moral rule of treating everyone with respect and to avoid causing harm. Virtue Ethics evaluates making decisions based on the virtues that a person believes in. This framework will

assess if the decisions that are made by engineers, exporters, and policymakers are ethical and virtuous. It emphasizes making decisions guided by virtues like honesty, responsibility, and compassion.

One of the rules that engineers must follow, “Engineers shall hold paramount the safety, health, and welfare of the public” (National Society of Professional Engineers, 2019). The responsibilities of engineers are “to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment” (Institute of Electrical and Electronics Engineers, 2020). These codes will be used to assess how engineers should act when involved in the product design of these devices. They also reinforce the idea that engineers have a duty to design responsibly, communicate truthfully, and anticipate long-term impacts.

The practice of exporting tons of electronic waste from developed countries like the US to developing countries has several harmful consequences including environmental pollution and health risks for workers and local populations. The biggest benefits of exporting e waste to developing countries are that it leads to cost savings for companies and removes the responsibility of disposing the waste within the developed country. But these benefits come at the cost of the human rights of the people living in the developing country. This waste causes harmful working conditions since it contains various hazardous substances that can harm the workers who process the e waste. “E-waste contains numerous toxic chemicals including metals such as lead, cadmium, mercury, and nickel, and organic compounds such as flame retardants, chlorofluorocarbons, polycyclic aromatic hydrocarbons (PAHs), polybrominated diphenyl ethers (PBDEs), and polychlorinated dibenzo-p-dioxins and furans (PCDD/Fs)” (Parvez, 2021). They can be exposed through inhalation of the toxic fumes, contact with the hazardous substances, and ingesting contaminated food and water. This waste also pollutes the soil, water, and air affecting the quality of life of the people that live close by. We will analyze this case using the two ethical frameworks that we mentioned previously: Deontological ethics and Virtue ethics.

Deontological Ethics emphasizes on following a set of moral rules/laws to guide decisions. From a deontological perspective the actions of causing harm and exploiting workers of another country are morally wrong. Exporting hazardous e waste to developing countries that are not capable of safely disposing it violates a few moral duties including: to avoid causing harm and to treat others as ends in themselves, not as a means to an end. The policy makers would receive some of the blame since they are unable to create or enforce stricter laws that reduce the amount of e waste that is exported to developing countries. The corporations and engineers that contribute to these exports are violating moral laws and that is deemed unethical from a deontological perspective.

Virtue Ethics emphasizes on following one’s own virtues to guide decisions. From a virtue ethics perspective, the decisions made by the companies and the engineers do not demonstrate virtues like honesty, responsibility, and compassion. The engineers involved in the product design create devices that have a limited lifespan and are hard to repair. They contribute to the concept of planned obsolescence. It is when businesses design products that are designed to have a limited lifespan, use low quality materials and parts that are more prone to failure, make it very difficult to repair the products, and stop software support. They do this to force consumers to purchase newer products and increase profits for the businesses. Contributing to this practice would increase the number of devices that consumers would have

to throw away. A virtuous engineer would not contribute to the practice of planned obsolescence, and they would design devices that have a longer lifespan and would be easy to repair. They would do this to save costs for consumers and reduce the number of devices that are thrown out each year. Contributing to planned obsolescence directly contributes to the problem of exporting hazardous e waste to developing countries and would be seen as unethical from a virtue ethics perspective.

Considering that we have deemed the practice of exporting tons of hazardous e waste to developing countries as unethical from a deontological and virtue ethics perspective, we must consider three options to resolve this case. The first option is to regulate or ban the export of e waste through creating and enforcing stricter laws within the developed countries. This option is deemed as ethical from a deontological perspective since it reduces harm and exploitation. This option is deemed as ethical from a virtue ethics perspective since it demonstrates good virtues like responsibility and compassion. The second option is to require electronics manufacturers to create sustainable and repairable electronics. This option is deemed as ethical from a deontological perspective since it emphasizes taking responsibility and reduces harm. This option is deemed as ethical from a virtue ethics perspective since it demonstrates good virtues like integrity and responsibility. The third option is to make electronics manufacturers take full responsibility for the lifecycle of their products including disposal by mandating Global Extended Producer Responsibility Programs (Source Intelligence, 2024). This option is ethical from a deontological perspective since it makes the corporations own the consequences of their own product designs. This option is ethical from a virtue ethics perspective since it demonstrates good virtues like responsibility and foresight.

Taking all these options into account, we will choose the one that makes that provides the best outcome. The most ethical option according to deontological and virtue ethics is to regulate or ban the export of e waste through creating and enforcing stricter laws within the developed countries. This option is supported from a deontological perspective since policy makers and corporations have a moral duty to not cause harm or exploit people from another country for the sake of convenience or profit. So, creating laws that can minimize harm and exploitation is ethical from a deontological perspective. Virtue ethics further supports this option since corporations, engineers, and policy makers should make decisions guided by virtues like integrity, responsibility, compassion, and foresight. Creating laws that consider the needs of people from another country and prevent local populations from being exposed to hazardous substances are made with good virtues in mind.

Comparing this option to the other options we can still see that it is the most advantageous one. For example, the option that requires electronic manufacturers to create sustainable and repairable electronics is a realistic and feasible solution, but it is not enough on its own to significantly reduce the amount of e waste that is being exported to developing countries. For this solution to be effective we would also have to implement other policies. The option that makes electronics manufacturers take full responsibility for the lifecycle of their products including disposal by mandating Global Extended Producer Responsibility Programs sounds morally right but is unrealistic. It would be too complex and take too long to implement. It would be difficult to have other countries cooperate in a global agreement since it might not be in their best interest.

The advantages of choosing the first option are that it is feasible when creating strict domestic laws and when signing international treaties. It would have an immediate impact in the reduction of e waste that is exported, and it would encourage safe domestic recycling. The disadvantages of choosing the first option are that it might be tough for certain countries like the US to implement these laws since they will have political pushback from the corporations that benefit from this practice. Another disadvantage is that there will be increased costs for the developed countries since they will have to invest in domestic recycling.

We rely on technology to complete many tasks and most consumers own more than one device to complete these tasks. It is important for consumers to know their options when it comes to getting rid of their devices when they are no longer needed. Consumers should also know the impacts of discarding their device when it is not properly recycled. This can help reduce the amount of e waste that is exported. This alongside with our solution of creating and enforcing stricter laws on exporting e waste to developing countries, we will be able to improve the quality of life of the people that live in developing countries. With reduced waste there will exist less pollution and it will prevent numerous health issues caused by exposure to hazardous substances. This solution is something that should be implemented in the US since they are the leading exporter of e-waste in the world.

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